Rumour Detection on Twitter Cascades: A Graph Neural Network Approach with Topological and XAI Analysis

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Abstract. Automated rumour detection on social media requires classifiers that generalise across news events, yet most prior work evaluates within a single train/test split. We study this on the PHEME dataset of 6,178 Twitter cascade graphs spanning seven real-world events, using a stratified 10% held-out test split that is fixed before hyperparameter search begins. We benchmark MLP, GCN, GAT, and GIN under three feature ablations (full, text-only, structural-only), and complement the classification study with a network science characterisation of cascade topology and GNNExplainer and GAT attention interpretability analyses. All four structural-only models cluster near the random lower bound (≈ 0.394), with F1 ranging from 0.484 (MLP) to 0.519 (GAT), confirming that pure topology provides only weak cross-event signal despite consistent structural differences between classes (rank-biserial $r < 0.15$ throughout). Text-based models generalise well; GIN with full features achieves the best F1 of 0.826, outperforming flat mean-pooled MLP by 0.081. Adding structural node features consistently hurts GCN and GAT but benefits GIN, suggesting sum aggregation can exploit degree distributions that mean aggregation cannot. GNNExplainer reveals that authority signals (follower count, PageRank) drive rumour predictions, while GAT attention concentrates on fewer nodes for rumours, consistent with an anomalous-node detection pattern.

1 Introduction

Misinformation on social media poses a challenge that scales with platform size. Vosoughi et al. [11] found that false news propagates roughly six times faster than true news on Twitter, reaching far larger audiences before corrections can circulate. The consequences, from electoral interference to vaccine hesitancy, have sustained a substantial body of automated detection research.

A Twitter reply thread is a natural directed graph: the source tweet is the root and each reply connects to its parent, forming a cascade tree that encodes both content and propagation dynamics. Most early detection systems treat rumour classification as a flat text problem, aggregating per-tweet features into a single vector and discarding the relational structure. Graph Neural Networks (GNNs) [5, 10] make it possible to learn directly over these cascade graphs. Prior work, notably BiGCN [2] (which propagates information bidirectionally through the cascade tree) and geometric deep learning on propagation graphs [8], has shown that exploiting graph structure improves over flat baselines. However, these studies typically evaluate within a single train/test split, leaving open whether GNNs learn event-transferable cascade signatures or whether apparent gains reflect overfitting to training-event topology distributions.

This paper addresses three questions using the PHEME dataset [14], a collection of 6,425 Twitter cascade graphs across nine real-world news events labelled as Rumour or Non-Rumour. First, does graph topology alone generalise across unseen events? Second, does message passing add value beyond mean-pooled text embeddings when node features are held constant? Third, which features drive GNN predictions, and do these patterns differ between classes? We adopt a stratified 10% held-out test split that is fixed before hyperparameter search, so the test set is never seen during model selection.

Our contributions are: (i) a topological characterisation across six network science analyses, finding consistently small structural differences between classes; (ii) a benchmark of MLP, GCN, GAT, and GIN under three feature ablations, with GIN achieving the best F1 of 0.826; and (iii) GNNExplainer and GAT attention analyses that identify degree centrality, follower count, and PageRank as the primary discriminative signals.

2 Dataset and Problem Formulation

2.1 The PHEME Dataset

We formulate rumour detection as binary graph-level classification. Given a cascade $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ (nodes = tweets, edges = replies, $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ the node feature matrix), the goal is to predict $y \in \{0, 1\}$ (Non-Rumour / Rumour) for the graph as a whole.

We use the PHEME dataset [14], comprising nine real-world Twitter news events with thread-level Rumour/Non-Rumour annotations. Two events are excluded due to near-degenerate class distributions: *ebola-essien* (100% Rumour, 14 cascades) and *prince-toronto* (98.3% Rumour, 233 cascades). The remaining seven events yield 6,178 graphs in total; event-level statistics are shown in Table 1. The positive rate varies from 22.0% (charliehebdo) to 52.9% (putinmissing), reflecting genuine differences in how each news event unfolded on Twitter. We apply a minimum-size filter of $|\mathcal{V}| \geq 3$, removing cascades that consist only of a source tweet and at most one reply.

Class imbalance is handled at training time by per-event majority-class undersampling, ensuring that no single event's distribution dominates the training set.

We release the dataset as a PyTorch Geometric archive on Zenodo [1] and Hugging Face.[1]

[1] `https://huggingface.co/datasets/NunoBatista/PHEME-Misinformation-Graphs`

Table 1. PHEME event statistics after excluding *ebola-essien* and *prince-toronto*. All seven events contribute to both hyperparameter search and final evaluation via a stratified 10% held-out split.

Event	Graphs	Rumour	Non-Rumour	R%
charliehebdo	2,079	458	1,621	22.0
ferguson	1,143	284	859	24.8
germanwings	469	238	231	50.7
gurlitt	138	61	77	44.2
ottawashooting	890	470	420	52.8
putinmissing	238	126	112	52.9
sydneysiege	1,221	522	699	42.8
Total	**6,178**	**2,159**	**4,019**	**34.9**

2.2 Graph Construction and Node Features

Graph construction. We parse each PHEME thread directory into a directed graph: the source tweet becomes the root node (index 0), and each reaction tweet adds a directed edge from its parent (identified via `in_reply_to_status_id`) to itself. Reactions whose parent is absent from the cascade are connected directly to the root, a common artefact of Twitter data collection. The result is a rooted directed tree where edges point from parent to child.

Node features. Each node carries a $d = 390$-dimensional feature vector: 384-dim text embeddings from `all-MiniLM-L6-v2` [9], 2 user features (log(1+followers) and verified status), and 4 structural features (PageRank, degree, in-degree, out-degree centrality). Two positional features, `is_root` and BFS `depth`, are appended at load time, bringing the full dimension to $d = 392$. We evaluate three feature configurations: *full* ($d = 392$), *text-only* ($d = 386$, NLP + user features), and *structural-only* ($d = 8$, user + graph-structural + positional features, no NLP).

2.3 Evaluation Protocol

We hold out a stratified 10% of each event's cascades as a fixed test set, sampling the Rumour and Non-Rumour classes independently within each event to preserve local class ratios. The held-out thread identifiers (≈ 540 graphs) are fixed before any model training, so the test set is never seen during hyperparameter search. Hyperparameters are selected on the remaining 90% using 5-fold stratified cross-validation (50 guided trials per model-configuration pair, macro F1 as the criterion). Final models are trained on the full 90% and evaluated on the held-out set. We report macro F1 (which weights both classes equally, penalising models that ignore the minority rumour class) and accuracy with bootstrap 95% confidence intervals (2,000 resamples).

3 Methodology

3.1 Baseline Model

Multi-Layer Perceptron (MLP). Each cascade is reduced to a single vector by mean-pooling its node feature matrix: $\mathbf{g} = \frac{1}{|\mathcal{V}|}\sum_v \mathbf{x}_v$. Fully connected layers with ReLU activations and dropout map this to a single output logit. Any GNN that outperforms the MLP gains something specifically from message passing rather than from feature richness alone.

3.2 Graph Neural Network Models

Graph Convolutional Network (GCN). We use the spectral graph convolution of Kipf and Welling [5]. At each layer ℓ, the representation of node v is updated as:

$$\tilde{\mathbf{h}}_v^{(\ell)} = \sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{1}{\sqrt{\tilde{d}_v}\sqrt{\tilde{d}_u}} \mathbf{h}_u^{(\ell-1)} \mathbf{W}^{(\ell)} \tag{1}$$

where $\tilde{d}_v = |\mathcal{N}(v)| + 1$ is the degree after adding self-loops. Pre-activations pass through a residual block with LayerNorm, stabilising training on the shallow cascades common in PHEME. Edge direction is a discrete hyperparameter for GCN and GAT: *original* (parent→child), *inverted* (child→parent, propagating leaf engagement signals toward the root), or *bidirectional* (both).

Graph Attention Network (GAT). We use the GATv2 attention mechanism of Brody et al. [3], which addresses a rank-collapse limitation of the original GAT [10] by computing a fully dynamic attention score:

$$e_{ij} = \mathbf{a}^\top \mathrm{LeakyReLU}\Big(\mathbf{W}\,[\mathbf{h}_i^{(\ell-1)} \,\|\, \mathbf{h}_j^{(\ell-1)}]\Big) \tag{2}$$

Scores are normalised by softmax over each node's neighbourhood. With K attention heads, representations are concatenated and passed through the same residual block as the GCN. The number of heads K is tuned during hyperparameter search.

Graph Isomorphism Network (GIN). GIN [12] applies a learnable-ϵ sum aggregation followed by an MLP per layer. Unlike mean pooling, sum preserves neighbour counts, making GIN as expressive as the Weisfeiler-Lehman test. GIN uses LayerNorm per layer without residual skips.

Shared readout. After L message-passing layers, all three models produce a graph-level representation by concatenating the mean-pooled node embeddings with the root node's embedding (preserving the source tweet's signal, which pooling alone would otherwise dilute):

$$\hat{\mathbf{g}} = \left[\frac{1}{|\mathcal{V}|}\sum_{v \in \mathcal{V}} \mathbf{h}_v^{(L)} \,\Big\|\, \mathbf{h}_0^{(L)}\right] \tag{3}$$

A single linear layer maps $\hat{\mathbf{g}}$ to the output logit: $\hat{y} = \sigma(\mathbf{w}_c^\top \hat{\mathbf{g}})$.

3.3 Training Procedure

Optimiser and schedule. All neural models use Adam with cosine annealing ($\eta_{\min} = \eta_0/10$) and gradient clipping (ℓ_2 norm 1.0).

Loss function. We use binary focal loss [6]:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}(1 - p_{t,i})^{\gamma} \log p_{t,i} \tag{4}$$

where $p_{t,i} = \sigma(\hat{y}_i)$ for positive examples and $1 - \sigma(\hat{y}_i)$ for negative ones; the modulating factor $(1 - p_{t,i})^{\gamma}$ down-weights easy examples and focuses training on hard cases.

Class balancing. The majority class is undersampled to match the minority class within each event, preventing skewed events from dominating the training signal.

Hyperparameter search. Hyperparameters are tuned via 50 guided trials per model-configuration pair (12 total), each evaluated by 5-fold stratified CV on the 90% pool (macro F1 criterion). The search space covers hidden dimensions, dropout, learning rate, weight decay, focal γ, attention heads (GAT only), and edge direction; the best configuration per pair is used for all final evaluations.

4 Related Work

GNN-based rumour and fake news detection. Graph-based approaches to rumour detection represent information cascades as graphs and apply message-passing to exploit propagation structure. Bian et al. [2] introduced BiGCN, which propagates representations bidirectionally along the cascade tree (top-down and bottom-up), consistently outperforming sequential baselines on Twitter15/16. Lu and Li [7] proposed GCAN, which jointly models user interaction graphs and source-content co-attention, also providing post-hoc explanations via attention weights. Monti et al. [8] applied spectral graph convolutions to the propagation graphs of FakeNewsNet, demonstrating that geometric deep learning transfers across datasets. Our study differs in using a held-out split fixed before hyperparameter search, performing a controlled feature ablation that isolates message passing from text richness, and complementing classification with topological analysis.

Explainability and cascade topology. GNNExplainer [13] provides post-hoc explanations by optimising a soft mask over the node features and subgraph of each prediction; we use it to compare the feature-group importance the model assigns to rumour versus non-rumour graphs. For cascade topology, Goel et al. [4] defined structural virality as the average pairwise distance in a diffusion tree, providing a principled measure of broadcast versus chain-like spread that we adopt in our topological analysis.

5 Network Science Analysis

We characterise structural properties of the full 6,425-graph dataset (all events, no size filter) using two-sided Mann-Whitney U tests. Rank-biserial r measures effect size; Spearman ρ measures correlation with the Rumour label for continuous metrics. Rumour cascades are consistently shallower and more broadcast-oriented, but all effects are small ($r < 0.15$) and do not generalise to classification (Section 6).

Cascade size and depth. Rumour cascades are significantly smaller (mean 13.89 nodes vs. 18.00; Spearman $\rho = -0.119$, $p < 0.001$) and shallower (mean depth 1.257 vs. 1.528; $\rho = -0.102$, $p < 0.001$). Stratifying by size reveals a tipping-point pattern: rumour prevalence declines monotonically from 44.9% among 1–5-node cascades to 11.1% above 200 nodes, consistent with factual content continuing to accumulate replies while rumour threads stall early.

Structural virality. Structural virality (average pairwise path length [4]) is significantly lower for rumour cascades (mean 2.463 vs. 2.808, $p < 0.001$, $r = 0.114$), confirming a more star-like topology; non-rumour cascades form longer diffusion chains.

Network robustness and echo chambers. Robustness is measured by tracking the largest connected component fraction after removing the top 1%–30% of degree-ranked hubs. Both classes show near-identical curves, differing by at most 0.014 at any removal rate. The branching factor (mean out-degree of internal nodes) is also indistinguishable between classes ($p = 0.650$, $r = 0.007$).

Further topological metrics. Table 2 reports six cascade complexity metrics. Five discriminate significantly. Root Dominance (fraction of all edges incident on the root node) is higher for rumours ($r = 0.074$), reinforcing the broadcast/star pattern. Strahler Number, a recursive measure of tree branching complexity analogous to stream order in hydrology, is lower for rumours ($r = 0.104$), indicating structurally simpler cascade trees. Normalised Depth (max depth divided by $\log_2 n$) is lower for rumours ($r = 0.070$), showing that rumour cascades are shallower than expected for their size. The Gini coefficient of the out-degree distribution ($0 =$ all nodes have equal out-degree; $1 =$ one node holds all edges) is lower for rumours ($r = 0.065$), meaning non-rumour cascades contain more high-degree hub nodes. Leaf Ratio (fraction of nodes with out-degree zero, i.e., no replies) also reaches significance ($p = 0.010$, $r = 0.039$). Epidemic R_0, the mean out-degree of nodes that themselves received at least one reply (analogous to the basic reproduction number in epidemiology), does not reach significance ($p = 0.081$).

Table 2. Topological metrics comparing Rumour (R) and Non-Rumour (NR) cascades (n = 6,425, Mann-Whitney U). r = rank-biserial effect size. Significance: *** $p < 0.001$, ** $p < 0.01$, ns $p \geq 0.05$.

Metric	R mean	NR mean	p-value	r	
Root Dominance	0.765	0.727	< 0.001	0.074	***
Gini out-degree	0.683	0.718	< 0.001	0.065	***
Strahler Number	1.893	2.024	< 0.001	0.104	***
Leaf Ratio	0.736	0.722	0.010	0.039	**
Norm. Depth	0.726	0.822	< 0.001	0.070	***
Epidemic R_0	4.214	4.469	0.081	0.028	ns

Table 3. Final held-out evaluation on the stratified 10% test set (≈ 540 graphs). Δ = margin above the random lower bound (F1 ≈ 0.394). **Bold**: best result.

Model	Config	F1	Acc	Δ
GCN	text	0.816	0.863	+0.422
GAT	text	0.809	0.857	+0.415
GIN	text	0.808	0.861	+0.414
MLP	text	0.747	0.802	+0.353
GIN	full	**0.826**	0.869	+0.432
GAT	full	0.786	0.846	+0.392
GCN	full	0.784	0.839	+0.390
MLP	full	0.745	0.800	+0.351
GAT	struct	0.519	0.615	+0.125
GIN	struct	0.511	0.565	+0.117
GCN	struct	0.492	0.630	+0.098
MLP	struct	0.484	0.515	+0.090

6 Classification Results

We evaluate four models under three feature configurations on the ≈ 540-graph held-out test set. The random lower bound (predict-all-non-rumour) is macro F1 ≈ 0.394; Δ in Table 3 is the margin above this bound.

Structural-only models. All four structural-only models cluster tightly near the random lower bound (F1 0.484–0.519, Δ +0.090 to +0.125). No architecture obtains reliable cross-event discriminative signal from pure topology; the gains are modest and overlapping in confidence intervals.

Text and full-feature models. All eight text and full-feature models exceed the lower bound by large margins (+0.351 to +0.432). GIN with full features achieves the best F1 of 0.826. For GCN and GAT, text-only configurations outperform full (+0.032 and +0.023), indicating structural node features add noise under mean aggregation. GIN reverses this pattern: full features are best (+0.018), consistent with sum aggregation extracting signal from degree distributions that mean pooling averages away.

Message passing vs. MLP. GCN-text and GAT-text outperform MLP-text by +0.069 and +0.062 F1 respectively. Bootstrap 95% CIs support this: GCN-text [0.775, 0.856] and GAT-text [0.765, 0.851] both have point estimates (0.816, 0.809) that exceed MLP-text's CI upper bound [0.700, 0.791]. NLP embeddings nonetheless carry the bulk of discriminative signal: the text-only MLP (0.747) already exceeds all structural-only models by a wide margin.

7 XAI Analysis

To understand what drives the GNN predictions, we apply two complementary explainability methods on a separate training run using the charliehebdo event as the held-out test set (1,952 graphs: 436 Rumour, 1,516 Non-Rumour). The models share the `gnn_full`/`gat_full` hyperparameters from Section 6 but are retrained on this separate split. Because charliehebdo has the lowest rumour prevalence (22.0%, Table 1), absolute F1 is lower than in the main evaluation (GCN: 0.601, GAT: 0.592); the importance patterns we report are not affected by this performance gap.

GNNExplainer: feature group importance. GNNExplainer [13] optimises a soft per-node mask over 80 balanced test graphs (40 per class); mask weights are averaged within semantic feature groups and compared via Mann-Whitney U (Fig. 1). The GCN assigns significantly higher importance to every group for Rumour than Non-Rumour. The largest gaps are in degree centrality ($r = 0.468$), follower count ($r = 0.455$), and PageRank ($r = 0.442$); all groups except out-degree are significant ($p < 0.01$).

GAT attention by cascade depth. The second GAT layer's attention weights (Eq. 2) are aggregated by the BFS depth of the receiving node. Depth-1 nodes (direct replies to the source tweet) show the strongest class difference: Non-Rumour cascades receive significantly more depth-1 attention (NR = 0.416 vs. R = 0.392, $p < 0.001$, $r = 0.459$), suggesting richer direct engagement in factual threads. Depth-3 shows a modest reversal (R > NR, $p < 0.001$, $r = 0.185$); beyond depth 3 differences are small or non-significant.

Attention entropy. Per-graph attention entropy is significantly lower for Rumour cascades (mean 2.404 vs. 2.630, $p < 0.001$): the GAT concentrates on fewer nodes when predicting rumours, consistent with the GNNExplainer finding that predictions are driven by a small set of structurally prominent nodes.

8 Discussion

The feature ablation confirms text embeddings as the primary driver of cross-event generalisation. GCN and GAT benefit from dropping structural node features, while GIN reverses this pattern – sum aggregation can extract signal from

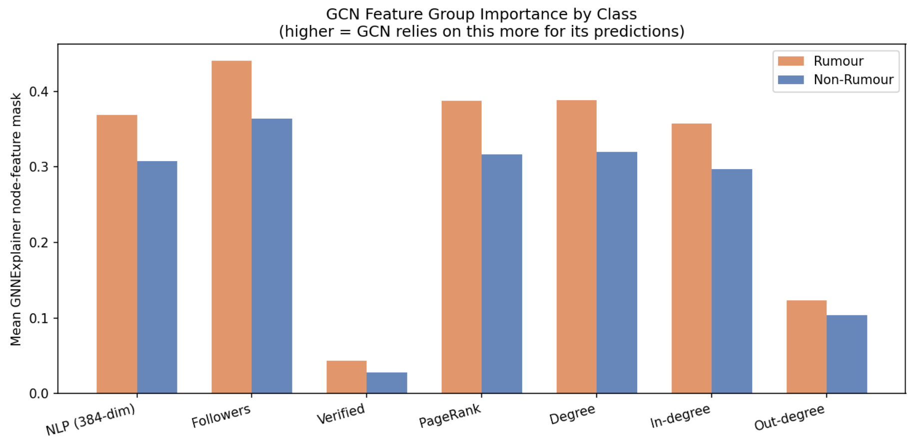

Fig. 1. GNNExplainer mean feature-group importance for Rumour and Non-Rumour graphs (80 test graphs, charliehebdo). Higher = relied upon more by the GCN. All groups except Out-degree differ significantly (Mann-Whitney $p < 0.01$).

PageRank and degree features that mean pooling averages away. All structural-only models cluster near the random lower bound (F1 0.48–0.52), indicating a weak but non-zero structural signal that no architecture reliably exploits for cross-event generalisation. The topological differences identified in Section 5 are consistent within events but appear unstable across them, which is why topology-only features cannot bridge the cross-event gap. Both XAI analyses converge on structural prominence as the primary discriminative signal: GNNExplainer assigns the largest importance gap to degree centrality, follower count, and PageRank, while GAT attention concentrates on fewer nodes for rumour cascades, consistent with anomalous-node detection.

The main limitation is single-dataset scope: PHEME covers English Twitter in 2014–2015, and cascade dynamics may differ across platforms and time periods. The stratified split is also sensitive to the random seed used for partitioning.

9 Conclusion

We benchmarked MLP, GCN, GAT, and GIN on 6,178 PHEME Twitter cascade graphs under a stratified held-out evaluation across seven events. Text embeddings dominate cross-event generalisation; topology alone provides weak, architecture-dependent signal. GIN with full features achieves the best F1 of 0.826, with GCN and GAT text-only models close behind (0.816, 0.809). GNNExplainer and GAT attention analyses reveal that authority signals drive rumour predictions and that attention concentrates on fewer nodes for rumours, consistent with anomalous-node detection. Future work should explore temporal posting dynamics, larger pre-trained encoders, and cross-platform evaluation.

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